SPE-226663-MS

Developments in Efforts to Replace Borate Crosslinked with High Viscosity Friction Reducers. Part II: Field Trials in Miqrat and Barik Sandstones

Hussein Al Salmi, BP, Muscat, Sultanate of Oman; Ahmed M. Gomaa and Alberto Casero, BP, Houston, TX, USA

Abstract

In the last 14 years, over 100 wells have been hydraulic fractured during the development phase of Khazzan and Ghazeer fields, Block 61, Oman. The primary stimulation approach consisted in massive single stage Tips Screen Out (TSO) hydraulic fracturing treatments with borate crosslinked (BXL) as fracturing fluid, also multistage verticals and horizontals have been applied in various occasions. While BXL fluids have been the standard fluid of choice for conventional fracturing for several decades because of their excellent proppant carrying capacity and minimal leak-off properties, reflected in the ability to generate large and conductive fractures, they also have a complex and fragile chemistry requiring specific attention to water quality and additive dosage, making their design and application a complex task. Additionally, it is well known and accepted and perhaps dismissed because of their routine application in the industry, that BXLF can cause considerable proppant pack and formation damage, especially in good quality formations typical of this sector, consequently affecting overall well performance.

To tackle this dilemma, the development of a more operationally friendly and less damaging substitute to BXL fluid is an ongoing industry effort that gained traction in recent years because of technological progress in fracturing fluid used in unconventional shale gas. This paper leverages this unconventional fracturing fluid technology progression and describes the field implementation of High Viscosity Friction Reducer (HVFR) fluid at high polymer concentration as a simpler to mix fluid, more stable at different water quality, and less damaging. Extensive laboratory process is described in Part I of this paper and was performed to properly formulate the fluid recipe and providing analogous performances to BXL fluid regarding viscosity, proppant transport and suspension capacity, and stability in high-temperature reservoir conditions.

This paper, Part II, describes the field implementation experiments in the Ghazeer field that confirmed an effective fluid and proppant placement, with leak off and fluid efficiency equivalent to standard BXL fluid, while improving operational and well performance. This paper examines the equipment readiness before field mobilization, results of frac diagnostic injections and frac treatments, the QAQC implemented on location, and the initial well-testing outcomes comparatively quantifying the formation damage linked to BXL fluid versus the modified HVFR.

The results underscore the capability of the novel fluid to enhance well performance and durability by reducing damage, representing substantial progress in hydraulic fracturing techniques for tight formations and the future way forward suggested for implementing conventional formations with higher permeability.

Introduction

The use of borate cross-linked gel has long been a standard in hydraulic fracturing, particularly for deep and high-temperature formations requiring high viscosity for proppant transport. However, these systems come with several challenges, including high chemical costs, potential formation damage due to residual polymers, and operational complexities related to crosslink timing and surface handling. In response, the industry has increasingly turned to HVFRs as a simpler and potentially more economical alternative, especially in unconventional and tight gas environments.

The search for more efficient and cost-effective fracturing fluids has led to increased interest in HVFRs as alternatives to traditional borate cross-linked gel systems. This paper presents the first field application of HVFR-based stimulation fluids in the Khazzan and Ghazeer fields, located in Block 61, Oman. These deep, high-pressure reservoirs (4500–5200 m depth, 7500–8500 psi formation pressure) are characterized by laminated, heterogeneous rocks that has historically posed significant stimulation challenges.

A structured field trial was designed to evaluate the performance of HVFRs under these demanding conditions. A surface mock test was first performed using different HVFR gpt loading to assess the surface equipment potential limitations and fluid viscosity build up then a data frac was performed using 30 gpt of HVFR over a single stage, successfully pumped at 60 bpm with no operational or surface issues. Following this, a full-scale frac stage was executed using 12 gpt HVFR with 100,000 lbs of 20/40 HSP proppant. The treatment was completed without anomalies, demonstrating fluid compatibility and effective proppant placement.

This paper discusses fluid selection rationale, treatment design, and execution results, highlighting the operational feasibility and performance of HVFR in a complex and deep gas formation. The early results suggest HVFRs may be a viable, cost-effective alternative to borate gel systems for conventional / unconventional tight gas reservoirs. The paper outlines the motivation for fluid replacement, treatment design, execution parameters, and results. It further explores how the use of HVFR can reduce complexity and cost, while maintaining effective proppant placement in deep gas reservoirs.

Barik and Miqrat Formation Description

Block 61 lies 275 km southwest of the capital of The Sultanate of Oman, Muscat. It contains the giant Paleozoic Khazzan and Ghazeer gas fields, which consist of two main reservoirs, Barik and Miqrat.

The Khazzan gas field in Block 61 was discovered in the early 1990s. In 2007, BP acquired the field and conducted an extensive appraisal program to support development plans for these challenging hydrocarbon accumulations (Rylance et al. 2011). The Khazzan project was endorsed in February 2014, which represented the first phase of development for the giant Khazzan Makarem stratigraphic structure (Fig. 1). In 2017, the Ghazeer field (a continuation of the Barik reservoir to the south and west) was merged with Block 61.

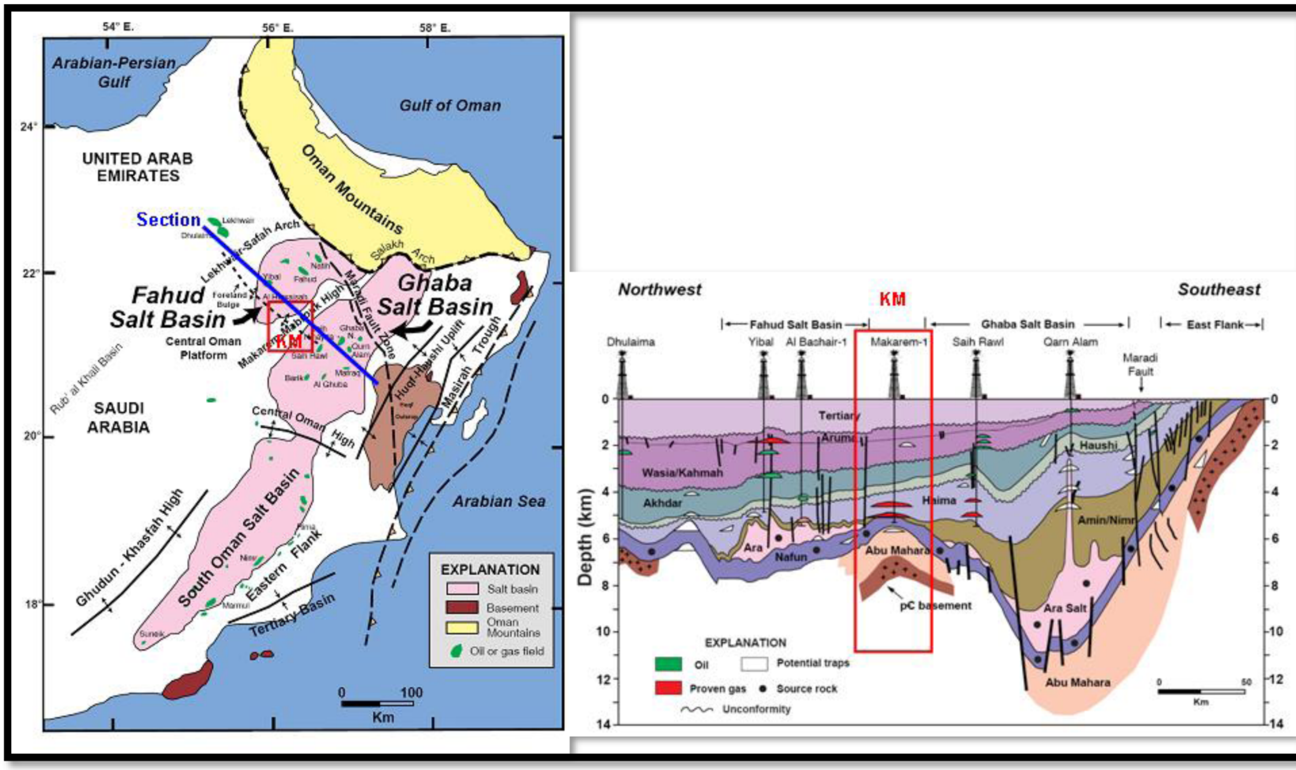

Figure 1—Major North Oman basins from Pollastro, 1999, modified from Loosveld et al., 1996, and a cross section through them, illustrating the KM high from Pollastro, 1999, modified from Droste, 1997.

As of 2021, Block 61 has been producing 1 Bcf/D from the Khazzan Barik reservoir since Sept 2017 and is producing an additional 0.5 Bcf/D from Ghazeer Barik reservoir since Oct 2020. The Miqrat reservoir is still under delineation. The Barik and Miqrat formations are located at depths of 4,200 - 4,800 m true vertical depth subsea (TVDSS) and all require a massive hydraulic fracturing stimulation to flow gas at economically attractive rates.

The Barik tight gas reservoir was appraised initially through a series of hydraulically fractured vertical and horizontal wells with a view to support the development of commingling gas production with the deeper Miqrat formation. The hydraulic fracture design applied in the vertical Barik completions was the adoption of massive conventional under-displaced BXL gel tip screen-out (TSO) treatments (Martins et al. 1989) with 0.5 to 1.0 million pounds of proppant placed, which consistently demonstrated skins of negative 6 or better (Clark et al, 2011). To date, more than 100 wells have been hydraulically fractured successfully with an outstanding success rate above 90%. Most of these wells are on production stream, which demonstrates the effectiveness of the hydraulic fracturing stimulation application to support producing the gas to surface.

Formation Geological Characteristics

The Barik Formation is Cambro-Ordovician in age and is part of the Haima Supergroup. It contains tight clastic, 80m thick reservoirs in Block 61. It produces sweet gas after hydraulic stimulation as its permeability is around 0.1 md, porosity around 6pu and its depth is around -4500m TVDSS. It forms a stratigraphic trap over the Makarem High by the northwards pinch-out of the fluvial to marginal marine deposits. The reservoir has been through various stages of diagenesis, which is the main controller of variations in reservoir quality along with sand thickness and vertical distance from the highly stressed mudstone units.

The Barik reservoir is part of the deep Haima gas play; it forms a stratigraphic trap within Block 61 due to the northward pinch-out of the fluvial to marginal marine Cambro-Ordovician sandstone over the Makarem (MKM) High (Al Rashdi and Spain 2015). The understanding of Barik geology within Block

61 has developed considerably during appraisal and early development phases of Khazzan. Over such a large area, the Barik demonstrates considerable variation in rock properties both vertically and areally; this variability is driven by depositional environment, lateral facies variability, and differences in diagenetic overprinting (Stiteler and Shell 2015).

Within Block 61, the Barik formation comprises alternating sharply delineated packages of sandstone and mudstone. Sequence stratigraphically, Al Bashair and Barik belong to a single fluvio-deltaic sequence that progrades across the subsiding platform margin from the south and southeast. The Barik sands are best developed in the south and pinch out to the north over the MKM structure (Figure 2 and Figure 3).

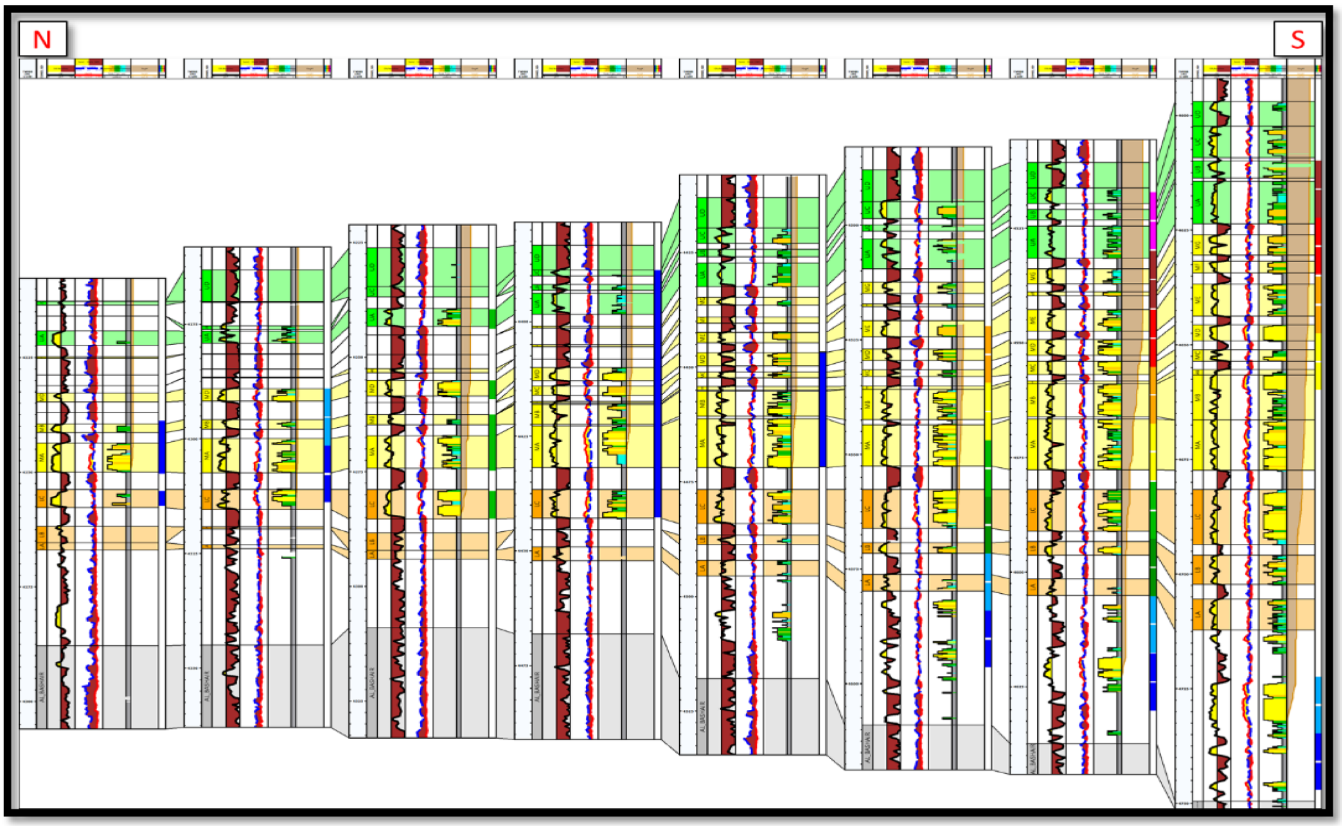

Figure 2—Barik correlation across the entire Block 61. Green corresponds to upper Barik, yellow corresponds to middle Barik and orange corresponds to lower Barik.

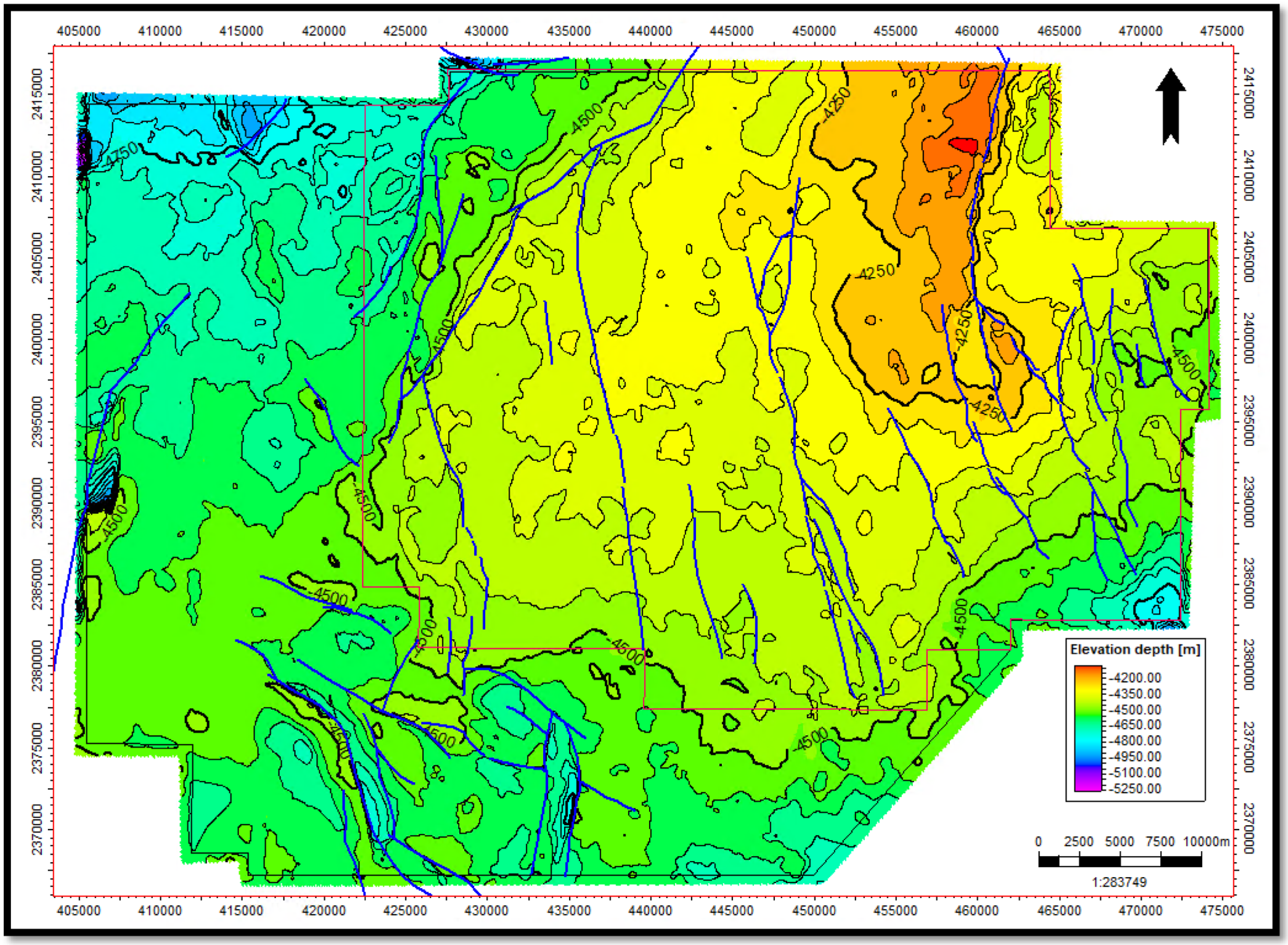

Figure 3—Barik top structure map with major faults (contour interval = 50 m).

The petrophysical description is a key component in predicting the reservoir's performance. The Barik petrophysical description has been significantly improved as the development phase has grown in Block 61. Since the start of the development phase, nearly all wells were logged with resistivity, neutron, density, and gamma ray, with a few exceptions due to operational limitations. Some key wells have been logged with special data acquisition programs to meet a particular objective or address a particular uncertainty. In addition, long sections of representative core have been acquired for the Barik and Miqrat formations. These data sets have been integrated to perform petrophysical evaluation, predict reservoir performance, improve hydraulic fracture models, and select perforation depths (Rylance et al, 2011; Spain et al. 2015). Figure 4 illustrates petrophysical logs for a typical Barik well.

The Miqrat Formation is Cambrian in age and is part of the Haima Supergroup. It contains tight & silty clastics, 50m thick Middle Miqrat reservoir in Block 61. It produces sweet gas after hydraulic stimulation as its permeability is less than 0.1 mD, porosity around 6pu and its depth is around -4700m TVDSS. It also forms a stratigraphic trap over the Makarem High by the northwards pinch-out of the alluvial plain to marginal marine deposits. The reservoir has been through various stages of diagenesis, which is the main controller of variations in reservoir quality along with sand minerology (chlorite & ductile). Compared to the Barik, the mudstones in Miqrat are less stressed and much easier to hydraulicly stimulate.

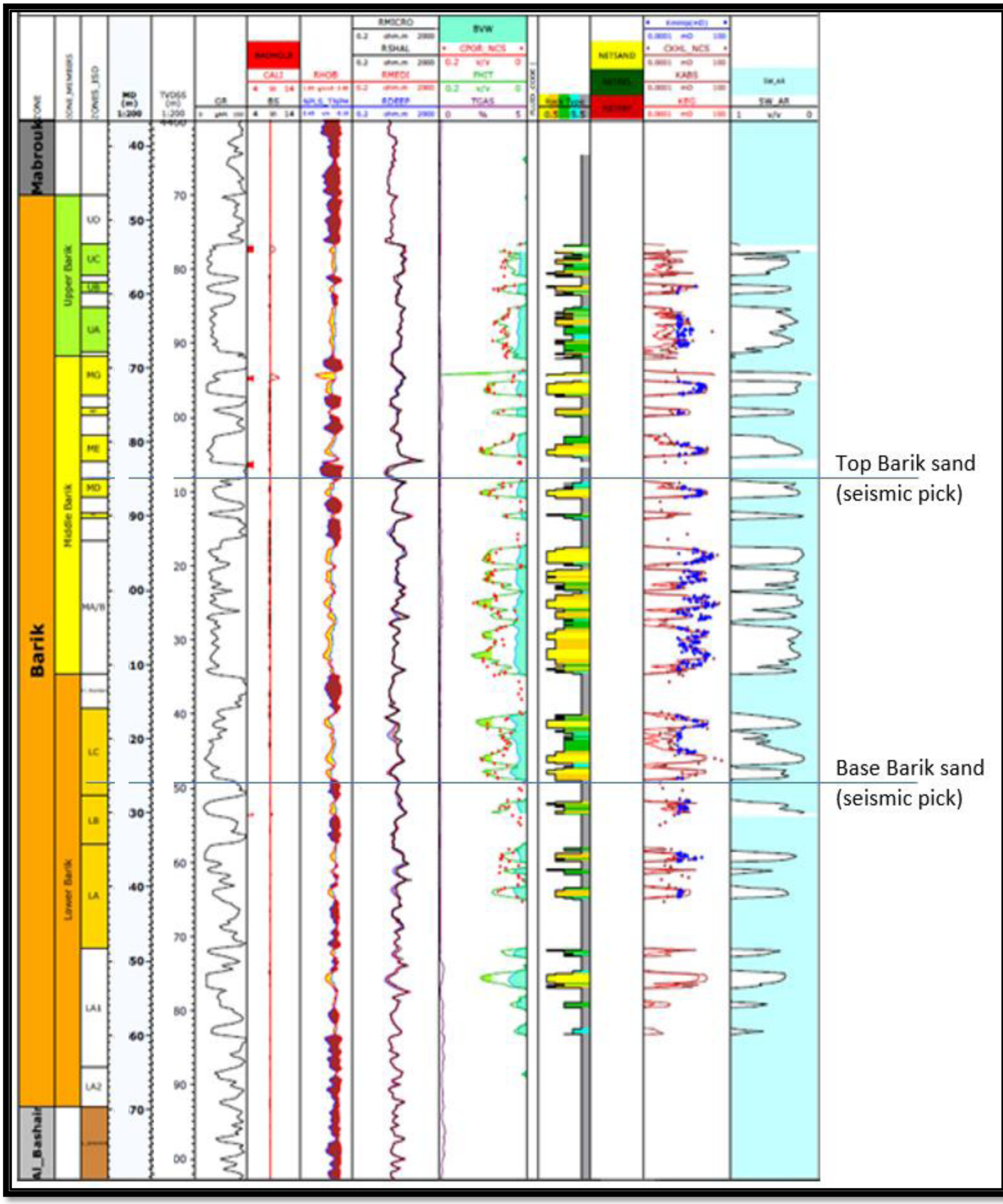

Figure 4—Barik type log showing Barik subdivisions and outlining seismic picks.

Barik Standard Fracturing Design

With a significantly average low permeability of 0.1 md or less, the Barik formation's tight gas reservoir requires hydraulic fracturing as an essential stimulation to improve the well productivity and optimize economic outcomes. The mudstone layers are distributed between the sandstones, and, in many cases, they add a set of boundaries that contribute to complexity in fracture growth and placement. This complex multilayering feature, combined with high stress contrast reflecting a complex stress regime and multiphase flow deriving from the valuable condensate hydrocarbon portion, presents a not trivial challenge for hydraulic fracturing placement and design, and ultimately well production performance.

The initial hydraulic fracturing design for the Barik formation followed a conventional approach, consisting of a single massive fracture intended to stimulate the entire pay zone in one treatment. This original design typically involved the placement of approximately one million pounds of a proppant blend (20/40 and 16/30 mesh high-strength proppant) across the full vertical extent of the reservoir interval. While this approach simplified execution and reduced stage count, post-job evaluations and production analyses revealed that it provided limited fracture coverage in the upper and lower segments of the formation, particularly in laminated or lower permeability layers.

To address these limitations, a comprehensive optimization study was undertaken, incorporating geological, petrophysical, and fracturing data. The study concluded that dividing the pay zone into multiple frac stages—targeting specific sections such as the lower, middle, and upper Barik intervals—would significantly enhance reservoir contact and improve hydrocarbon recovery. The refined design aimed to deliver more uniform fracture propagation, reduce stress shadowing effects, and ensure better proppant distribution across all reservoirs sub-layers. This evolution in the fracturing strategy allowed for improved stimulation efficiency and production uplift, particularly from zones that were previously under-stimulated in the single-stage design.

Methodology and Previous Work

A systematic methodology was followed to ensure a technically sound and operationally reliable approach for introducing HVFR fluid into the hydraulic fracturing program. The project commenced with laboratory testing to assess the fluid's thermal stability, viscosity performance, and formation compatibility, establishing its technical value. This was followed by a surface-based Mock Test to validate the fluid's mixability, pumpability, and interaction with field equipment at a concentration of 30 gpt. Upon successful surface validation, a data frac was conducted using 30 gpt HVFR to evaluate injection behavior, pressure trends, and fluid efficiency in the formation. Building on those results, a full-scale hydraulic fracturing stage was performed using a reduced HVFR loading of 12 gpt to assess proppant transport and downhole performance under real conditions. Following the success of the 12 gpt trial, a second full-scale hydraulic fracturing stage was executed at the original concentration of 30 gpt to further confirm fluid performance and treatment reliability at higher loading. This structured step-by-step process ensured a safe and confident transition from laboratory concept to full-field application. Summarizing the methodology followed the streamline below:

1. Lab Testing
2. Mock Field Trial
3. Datafrac (no proppant)
4. Full Job at low 12 gpt, introducing proppant
5. Full Job at design 30 gpt, with proppant

Laboratory Testing and validation

Laboratory testing (Gomaa et al., 2025) confirmed that the HVFR system exhibits stable viscosity performance at elevated temperatures, specifically at 290°F, making it a strong candidate for replacing traditional BXL gels in Khazzan and Ghazeer high-reservoir temperature conditions. At a concentration of 30 gpt, the HVFR fluid consistently achieved viscosities around 300 cP at a shear rate of 100 s^{-1}, which is sufficient to achieve effective proppant transport across the extended fracture network. Furthermore, increasing the concentration of HVFR did not show to enhance viscosity retention and thermal stability, providing confidence in the fluid's ability to maintain consistent rheological behavior under prolonged high-temperature exposure.

Breaker performance was also evaluated to determine optimal loading for efficient viscosity reduction post-treatment. Results showed that higher breaker concentrations led to faster and more complete viscosity degradation. Particularly, a breaker loading of 5 ppt achieved the most rapid and controlled viscosity decline, supporting effective fracture clean-up without compromising early-stage proppant transport. Compatibility tests with formation fluids demonstrated that the HVFR system maintained chemical stability in the presence of elevated iron concentrations, with no signs of emulsion formation, phase separation, or adverse reactions, highlighting its robustness in chemically complex downhole environments.

Fluid loss and leak-off control evaluations indicated that the HVFR system offers high fluid efficiency with minimal risk of formation damage. Unlike BXL gels, the HVFR did not generate a filter cake, reducing the potential for long-term permeability impairment near the fracture face. In proppant suspension tests, the HVFR fluid successfully kept proppant suspended for up to 25 minutes at 290°F with minimal settling, outperforming conventional BXL systems under equivalent test conditions.

Finally, conductivity recovery and clean-up performance further reinforced the HVFR fluid's advantages. Under high-temperature and high-pressure test conditions, the fluid maintained elevated fracture conductivity while minimizing residual damage. These results collectively demonstrate that the HVFR system provides substantial operational benefits, including improved formation compatibility, reduced damage potential, effective proppant transport, and reliable viscosity control—all of which are critical for maximizing stimulation efficiency and production in high-pressure, high-temperature gas reservoirs. The laboratory testing referenced in this section is well documented and presented in detail in SPE Paper No. SPE-226629, which supports the technical foundation for the field application results discussed in this paper.

Mock Test (Surface Evaluation)

Prior to field implementation, a mock test was conducted to evaluate the surface compatibility and operational handling of the HVFR d fluid under simulated field conditions. Unlike traditional lab-based compatibility tests, this surface equipment focused test aimed to assess real-time interactions of the fluid with mixing and pumping equipment and to identify any potential operational complications. Given the elastic and shear sensitivity nature of HVFR fluids, it was critical to confirm that the system could be mixed, transferred, and pumped at the required rates without causing operational issues. During the mock test, attention gibe to the following potential issues:

- Over-pressuring during blending,
- Fluid gelling in lines or Frac pumps
- Excessive foaming or separation,
- Mechanical strain or damage to surface equipment.

The mock test setup consisted of:

- Location Selection: conducted at field location using blender, hydration units, and high-pressure pumps intended for the field job with the same set up used during an actual frac job.
- Fluid Formulation: Full real job formulation, fluid mixed using the full job-intended chemistry, HVFR concentration, and additives (e.g., biocide, scale inhibitor, breaker, etc.).
- Volume Simulated: ~500 bbl equivalent recirculated.

The key objectives of the mock test were:

- Validate fluid mixing and hydration behavior in field-scale tanks.
- Monitor fluid flow through surface lines and valves.

- Check for anomalies in pressure, vibration, or flow restrictions.
- Assess fluid response at various shear rates typical of blender and pump conditions.

The mock test was executed with no signs of foaming, gelling, or separation were observed during or after hydration. The fluid remained stable and elastic, showing no indication of crosslink-like buildup or line obstruction. All surface equipment (blender, lines, valves, hydration units) was visually inspected post-test; no chemical fouling, erosion, or mechanical damage was found at the blender or the pumps. The pumping simulation confirmed stable pressure and flow, verifying the fluid's suitability for 60 bpm injection rates.

These results provided critical confidence in proceeding with the data frac and full-stage field application, ensuring that the elastic behavior of the HVFR would not impose additional risks or operational downtime on the surface system.

Data Frac Injection

To validate the operational and reservoir compatibility of the HVFR fluid under realistic field conditions, a data frac stage was executed as an intermediate step between lab qualification and full-frac deployment.

The key objectives of the datafrac injection were:

- Confirm that the HVFR fluid could be pumped at high pumping rates and polymer loading without friction-related issues or surface complications.
- Assess fluid efficiency in the reservoir, including potential differences in pressure response, leakoff, and closure behavior, and compare them to historical BXL jobs. Study the outcomes and operation learnings to build the case for full proppant treatments.

Design Parameters

The data frac was designed as a single-stage injection to evaluate HVFR performance at elevated concentrations. 30 gpt HVFR fluid was prepared and pumped at a rate of 40-45 barrels per minute (bpm), with a total volume of 500 barrels. The treatment was executed without proppant to isolate the fluid behavior and pressure response. The targeted interval was a laminated, heterogeneous gas-bearing zone located at approximately 5000 meters true vertical depth. The injection was closely monitored through surface treating pressure, instantaneous shut-in pressure (ISIP), and fall-off trends to assess fluid efficiency and compare fracture response against historical BXL gel treatments.

Operational Observations

The HVFR fluid was successfully pumped (Figure 5) at the full design rate of 40-45 bpm without any operational issues throughout the datafrac. Surface pressure remained stable, with no signs of foaming, gelling, or separation during the mixing or injection process. The elastic nature of the fluid did not create any pumping challenges, validating the findings from the earlier mock test. Equipment response was smooth, and no anomalies were observed in pressure behavior, confirming that the HVFR system could be reliably managed using the existing surface infrastructure under high-rate injection conditions.

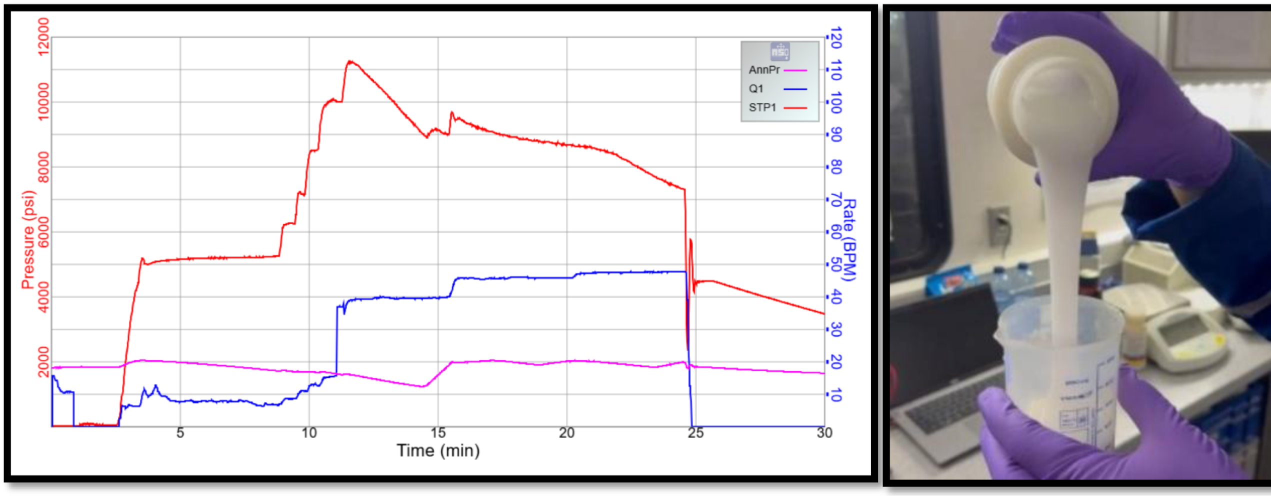

Figure 5—Datafrac injection with 30gpt HVFR fluid & fluid sample.

To assess the friction reduction performance of the HVFR fluid relative to conventional borate BXL systems, a comparative analysis was performed using surface treating pressure data from three data frac treatments. The comparison included the HVFR data frac using 30 gpt loading (G1) and two offset wells that were treated with 30 gpt of BXL fluid (G2&G3). Each of the treatments G1&G2 was executed with a total pumped volume of 500 barrels and 1000 barrels on G3, ensuring a consistent basis for evaluation. The pressure profiles were plotted to compare surface treating pressure versus time, focusing on the pressure drop behavior after the fluid front reached the perforations (Figure 6).

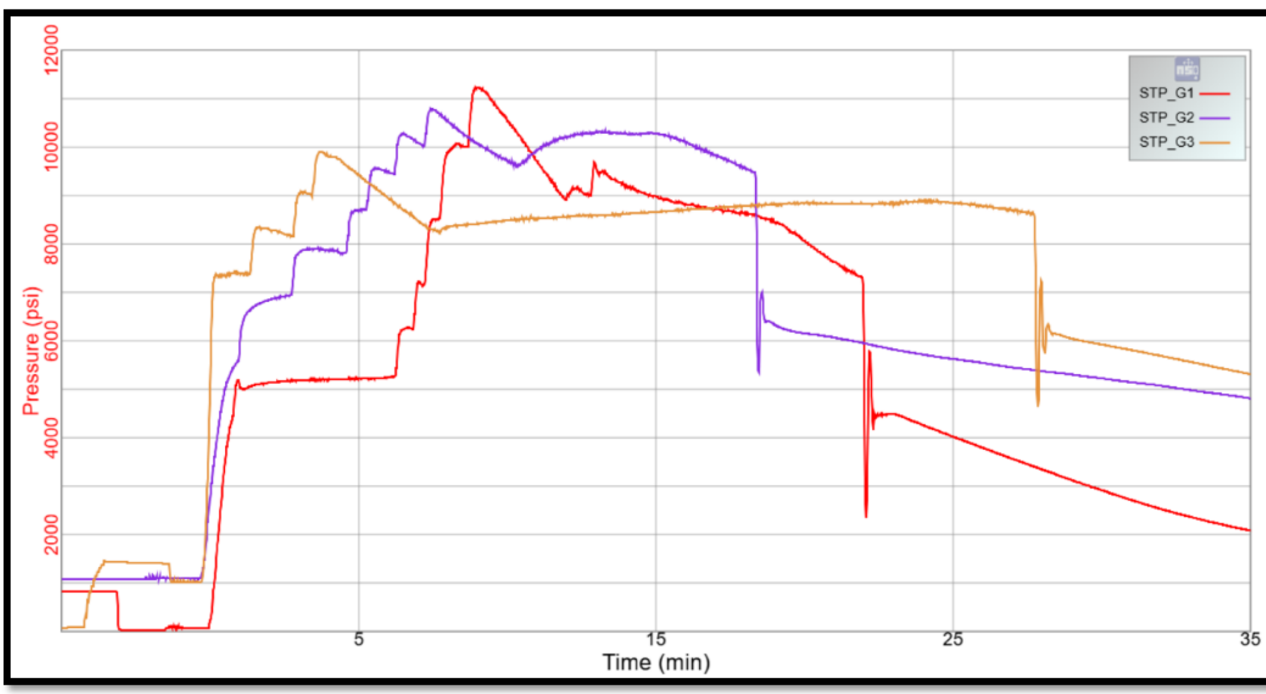

Figure 6—Datafrac injection surface treating pressure vs offsets

The results clearly indicate that the HVFR fluid exhibited a more pronounced and continuous pressure drop throughout the pumping period compared to the BXL fluids. This trend could reflects a superior friction reduction capability, as the HVFR maintained lower surface pressures under identical pumping conditions. While the BXL-treated wells showed relatively stable or fluctuating pressure profiles, the HVFR-treated stage demonstrated a steady pressure decline over time which could be attributed to the fluid ability to reduce friction losses more effectively during high-rate injection. These findings highlight the operational benefits of HVFR fluids in terms of energy efficiency and reduced surface pressure loading, which can contribute to safer and more cost-effective fracturing operations in high-pressure formations like the Barik & Miqrat.

The HVFR-treated well demonstrated a notable and continuous pressure drop of approximately 2120 psi over 8.6 minutes plot xxx, indicating strong friction reduction as the fluid moved through the wellbore and into the formation. In contrast, the BXL-treated offset wells exhibited only marginal pressure decline during the same injection window, with relatively stable or slightly fluctuating pressure profiles. These results are clearly presented in the comparative pressure plot (Figure 7), which highlights the distinct pressure reduction trend achieved with HVFR.

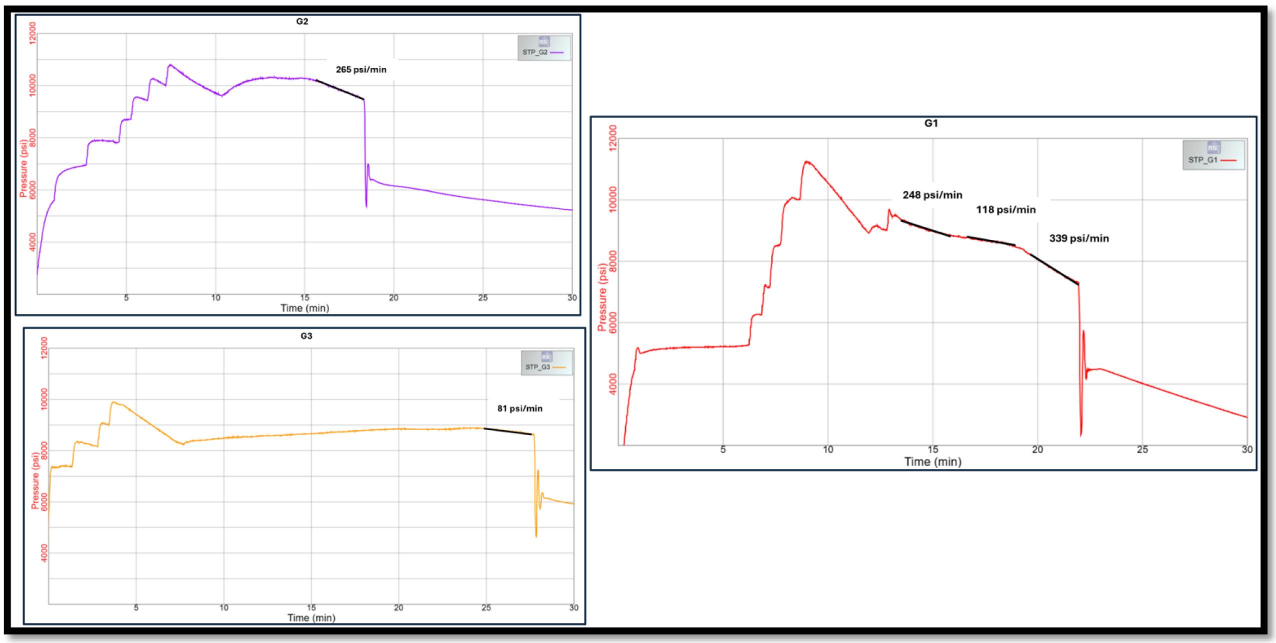

Figure 7—– surface treating pressure rate of decline while pumping HVFR in offset wells (left) and well treated with HVFR (right).

This behavior reinforces the friction reduction capability of the HVFR system, which not only lowers surface treating pressures but also enhances energy efficiency and operational safety. The continuous pressure decline with HVFR supports its value in high-rate pumping operations, particularly in deep, high-pressure formations like the Barik & Miqrat, where reducing frictional resistance can significantly impact overall stimulation efficiency and surface equipment demands.

The fluid efficiency results from the data frac injections were evaluated across three wells—G1, G2, and G3—yielding measured efficiencies of 24%, 47%, and 40%, respectively. The HVFR-treated well (G1) exhibited the lowest fluid efficiency at 24%, compared to higher values observed in the wells treated with borate BXL fluids. This difference may be attributed to formation-specific geological characteristics, rather than fluid performance limitations. The Barik formation, where these tests were conducted, is known for its highly laminated structure, with a complex network of natural features such as fissures, bedding planes, and vertical permeability variations. These characteristics contribute to enhanced fluid leakoff, especially in the absence of sealing mechanisms.

The HVFR fluid, which is designed to minimize residue and does not form a filter cake, likely allowed for more realistic leakoff into the natural features of the formation. In contrast, the BXL fluids have improved fluid efficiency due to filter cake formation, which restricted further fluid loss into the matrix or natural pathways. While this may seem advantageous in terms of fluid efficiency metrics, it can also lead to fracture face damage and impaired conductivity.

This observations highlight a key learning from the HVFR datafrac: in formations with complex leakoff behavior, fluid efficiency can potentially be improved using fluid loss additives (FLA). Incorporating FLA into the HVFR system could reduce uncontrolled fluid invasion, mitigate excessive leakoff, and help bridge micro-fractures or high-permeability streaks—ultimately enhancing fluid retention within the fracture and increasing stimulation effectiveness. The datafrac injection provided clear validation of the HVFR fluid ability to be pumped at high concentration without operational or surface equipment issues. The comparative analysis with offset wells treated using BXL fluids highlighted the HVFRs support the friction reduction performance and more efficient pressure response within the formation. These results not only demonstrated the technical value of HVFR over traditional BXL systems but also justified the progression to a full-scale fracturing stage using a reduced loading of 12 gpt, incorporating proppant to further assess the fluid transport capability on actual hydraulic fracture injection.

1st Full Field Trial (12 gpt HVFR)

Following the successful mock test and data frac injection, a full-scale hydraulic fracturing stage was executed using a reduced HVFR loading of 12 gpt (Table 1 and Figure 8). The objective of this trial was to evaluate the effectiveness of HVFR in carrying proppant and achieving fracture conductivity under realistic field conditions, while minimizing chemical loading and cost.

It is important to note that this fracturing treatment was intentionally designed as a hybrid concept, representing a significant departure from the conventional BXL gel frac designs typically used in the Barik formation. Traditionally, these treatments utilize a linear gel pad followed by a BXL gel during proppant stages. However, in this case, the objective was to implement a design that better promoted fracture length extension while reducing the risk of intersecting a nearby massive fracture that had recently been placed in the deeper section of the Middle Barik interval.

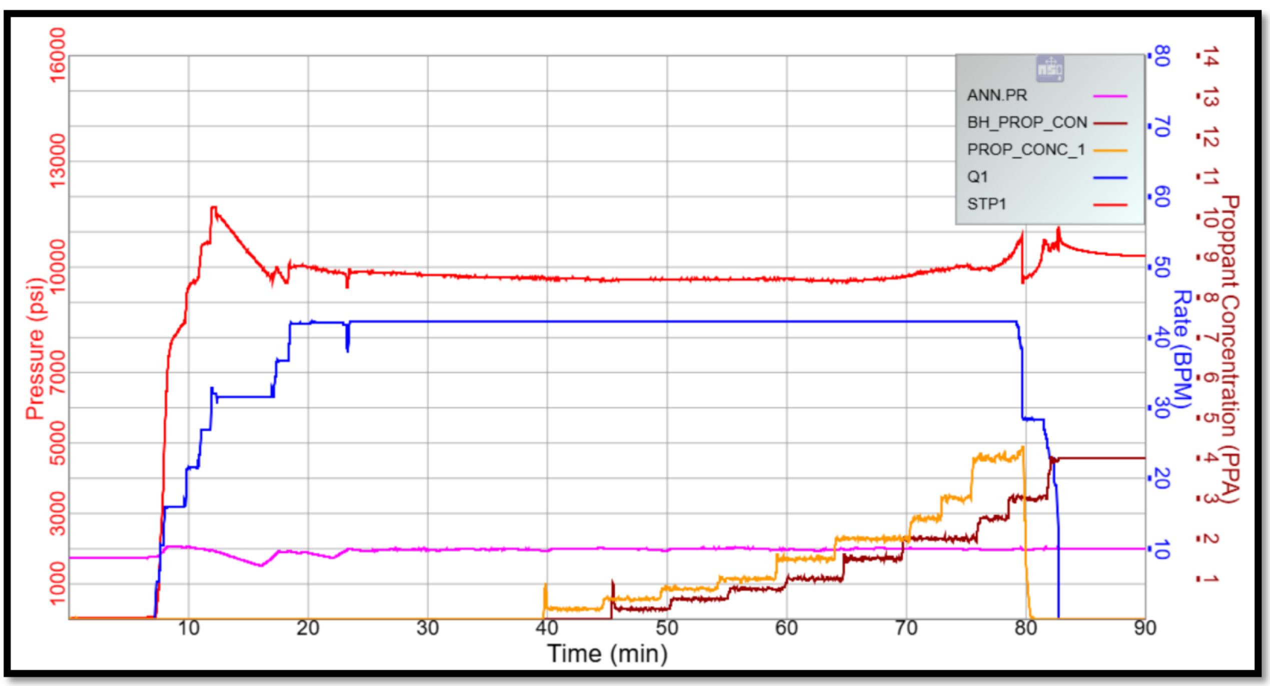

Figure 8—12 gpt HVFR proppant fracturing treatment chart.

Table 1—Pump Schedule - 1st full job field trial.

Step #	Step Name	Fluid Type	CFLD Vol bbl	Slurry Volume bbl	Slurry Rate bbl/min	Prop Type	Prop Conc PPA	Prop Mass lb	Pump Time min
1	PAD	HVFR-12 gpt	1200.0	1200	35.2	N/A	0.0	0	34.1
2	0.25 PPA	HVFR-12 gpt	188.0	201.7	42.3	20/40 HSP	0.2	2,120	4.8
3	0.5 PPA	HVFR-12 gpt	188.6	203.7	42.3	20/40 HSP	0.5	3,880	4.8
4	0.75 PPA	HVFR-12 gpt	188.7	205.3	42.3	20/40 HSP	0.7	5,834	4.9
5	1 PPA	HVFR-12 gpt	188.0	206.6	42.3	20/40 HSP	1.0	7,828	4.9
6	1.5 PPA	HVFR-12 gpt	188.6	210.3	42.3	20/40 HSP	1.5	11,723	5
7	2 PPA	HVFR-12 gpt	235.9	267.1	42.3	20/40 HSP	2.0	19,649	6.3
8	2.5 PPA	HVFR-12 gpt	94.5	108.7	42.3	20/40 HSP	2.5	9,722	2.6
9	3 PPA	HVFR-12 gpt	94.4	110.3	42.3	20/40 HSP	3.0	11,730	2.6
10	4 PPA	HVFR-12 gpt	184.3	217.4	39.5	20/40 HSP	4.0	27,394	5.5
11	Flush	HVFR-2 gpt	50.0	50	16.8	N/A	0.0	0	2.9

Given the lack of a direct analogue in previous Barik sandstone stimulation designs, this job stands out as a unique application. To reflect the hybrid nature of the design, a lower polymer concentration of 12 gallons per thousand (gpt) of HVFR was selected, significantly lower than the 30 gpt design used to match the viscosity of BXL gels. This lower loading was aimed at achieving fracture extension more akin to slickwater treatments, while still maintaining sufficient proppant transport capabilities.

The primary distinction between the hybrid BXL design and the HVFR treatment lies in the fluid used during the proppant stages. In a typical hybrid BXL job, the proppant-laden portion of the treatment would involve a switch from linear gel to BXL gel to ensure high viscosity for proppant carrying. In contrast, the HVFR treatment maintained the 12 gpt concentration throughout the entire job, including the proppant stages, maintaining a fluid profile more comparable to a slickwater approach. This strategy offered the dual benefit of minimizing formation damage while still delivering adequate proppant placement and fracture geometry control.

The treatment targeted the top section of the middle Barik formation, a laminated and heterogeneous interval located at approximately 5000 meters depth. A single-stage design was implemented, incorporating 100,000 pounds of 20/40 mesh high-strength proppant (HSP). The pumping schedule was tailored to maintain proppant suspension while delivering the job at the required rate, using the 12 gpt HVFR fluid as the sole carrier system.

The proppant pump schedule for the HVFR data frac was intentionally designed with a conservative ramp-up approach, as this represented the first field application of HVFR fluid for proppant transport in this formation. The targeted zone—located in the upper section of the middle Barik formation—was considered technically challenging. During the initial water injection (breakdown injection), surface treating pressures approached the upper threshold of the allowable pressure envelope, indicating that the treatment would be difficult to execute and closely constrained by surface equipment limitations.

Given these conditions, the design focused on gradually increasing the proppant concentration, allowing the operations team to monitor real-time well response, pressure behavior, and surface loads. Despite the elevated pressure environment, the HVFR system demonstrated stable pumpability and effective proppant-carrying performance. The job was executed according to the planned schedule, with 100% of the designed proppant volume (100,000 pounds) successfully pumped into the well. However, due to the high-pressure nature of the formation and dynamic injection conditions, only 75% of the proppant (approximately 75,000 pounds) was effectively placed into the fracture.

The job was terminated in a pressure out during the flush. This is likely due to the choice of using a 12gpt through all the job which essentially is a slickwater treatment in a formation that typically has proppant entry issue (i.e., frac narrowness and complexity). Nevertheless, the job resulted in a 70% placement and considering the low fluid viscosity, it is a remarkable achievement.

As shown in Figure 8, the treatment proceeded with high surface treating pressure. Achieving full fluid delivery and substantial proppant placement under such challenging pressure conditions further validated the HVFR fluid's capability to perform in high-stress reservoirs while providing valuable insights into optimizing future designs for similar environments.

Operationally, the job was executed without surface equipment issues. The treating pressure remained within the limitation envelope, and the fracture geometry developed in line with pre-job modeling. Post-treatment evaluation indicated effective proppant placement, suggesting that even at lower loading, the HVFR maintained sufficient viscosity and elasticity to ensure transport performance.

Fluid samples were collected during the treatment at both the clean fluid (pad) stages and the proppant-laden stages to validate key quality parameters. These samples were analyzed to confirm that the HVFR maintained the expected viscosity profile throughout the job and to ensure proper proppant suspension and uniform mixing within the fluid system. This real-time sampling provided assurance that the fluid preparation and blending processes remained consistent with the design specifications during execution Figure 9.

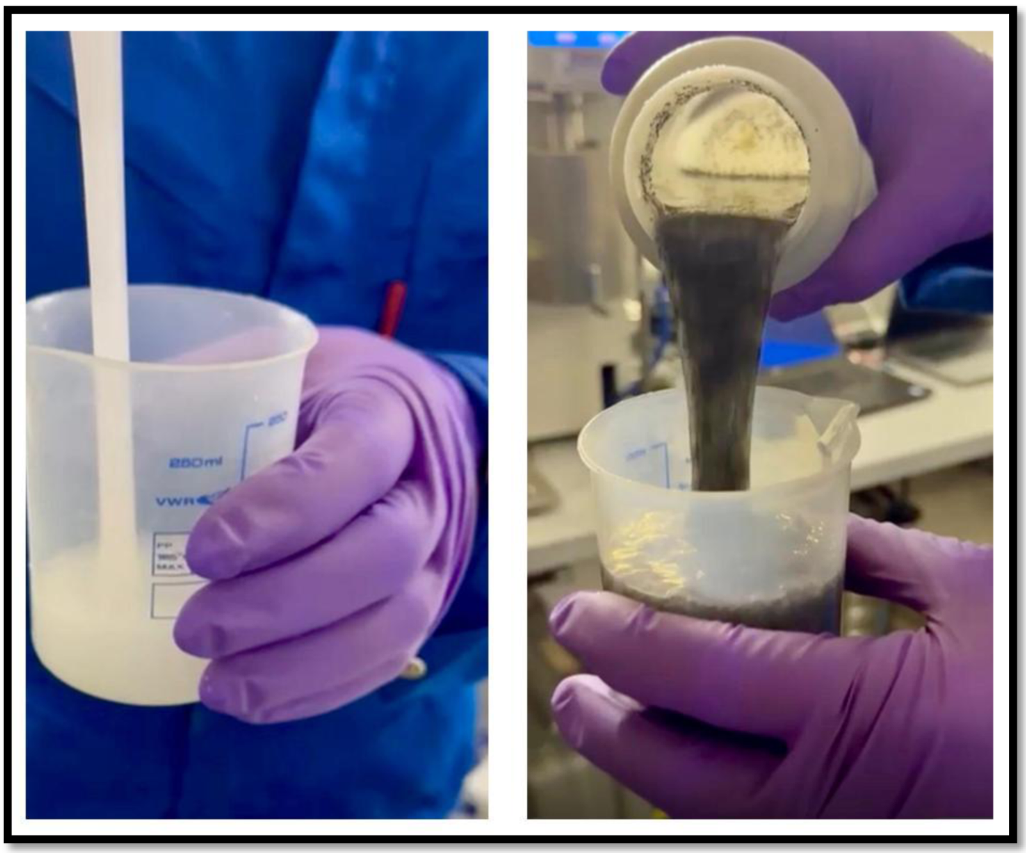

Figure 9—HVFR (12gpt) samples collected during 1st full job trial, without proppant (left and center) and with proppant (right).

This 12 gpt field trial not only validated the downhole behavior of HVFR under proppant-laden conditions but also demonstrated the fluid capacity to deliver operational reliability and stimulation quality comparable to conventional systems—at reduced chemical concentrations.

Conclusions

This paper presented a structured and stepwise approach for evaluating and implementing HVFR fluid as an alternative to BXL gel in the hydraulic fracturing of deep, high-pressure, high-temperature gas reservoirs within the Barik formation of Block 61, Oman. The methodology combined laboratory validation, surface-based mock testing, data frac calibration, and full-scale proppant trial stages to systematically de-risk the transition from conventional gel systems to HVFR-based designs.

Laboratory tests confirmed that the HVFR fluid maintained thermal stability and viscosity at 290°F and provided effective proppant suspension. The surface mock test validated that the elastic nature of the HVFR fluid did not cause foaming, gelling, or surface equipment issues under high-rate field conditions.

The initial data frac, executed at 30 gpt with clean fluid, demonstrated excellent friction reduction and pressure control, outperforming offset wells treated with XL fluid. Surface treating pressure declined steadily, with a total pressure reduction of over 2100 psi in under nine minutes. Although the HVFR-treated well showed lower fluid efficiency, this was attributed to natural leakoff pathways in the laminated Barik formation, rather than fluid design deficiencies. Future treatments may incorporate fluid loss additives to improve efficiency further.

A subsequent full frac at 12 gpt with 100,000 lbs of proppant targeted a geologically complex zone and was executed successfully despite high treating pressures near the upper operational limit. Approximately 75% of the planned proppant volume was placed, with 100% of the fluid volume delivered, validating the HVFR's capability to transport proppant under challenging conditions. Additional trials are planned and on-going when writing this paper.

Overall, the study demonstrates that HVFR fluid systems offer a viable, efficient, and lower-residue alternative to BXL gels for hydraulic fracturing in deep gas formations. Key benefits include simplified logistics, reduced surface pressure loading, minimized formation damage, and reliable proppant transport performance. The learnings from this multi-phase evaluation provide a foundation for broader implementation of HVFR technology in similar high-pressure, laminated formations.

Acknowledgement

The authors would like to thank BP for their support and permission to publish this paper. The authors would also like to thank all the colleagues who have contributed directly or indirectly to this work.

Units of Measurement

cP	Fluid viscosity, centipoise
ºF	Temperature, degree Fahrenheit
gpt	Liquid chemical concentration, gallon per thousand gallons of fluid
km	Distance, kilometer
km^2	Area, square kilometer
lb/ft^2	Proppant concentration in the fracture, Pound per square foot
m	Distance, meter
md	Permeability, Darcy x 10^{-3}
MDa	Molecular weight, mega-dalton
mesh	Diameter, wires per inch.
min	Time, minutes
ml/min	Leak-off rate, milliliter per minute
PPA	Proppant stage concentration, pounds mass of proppant added in a gallon of clean fluid.
ppm	concentration, parts per million
ppt	Dry chemical concentration, pound per thousand gal of fluid
psi	Pressure, pound force per square inch

rpm Speed, revolutions per minute
s^{-1} Shear rate, reciprocal seconds.
Concentration, guar polymer, in lbs/1000gal

Nomenclature

BXL Borate crosslinked fluid
frac Fracture
HSP High Strength Proppant
HTHP High Temperature High Pressure
HVFR High Viscosity Friction Reducer
LB Lower Barik
MB Middle Barik
TDS Total Dissolved Solids
TSO Tip Screen Out
UB Upper Barik

References

Al Rashdi, K. and Spain, D. 2015. Barik Reservoir Description – An Evolving Concept. Paper presented at the SPE Middle East Unconventional Resources Conference and Exhibition, Muscat, Oman, 26–28 January. SPE-172912-MS. https://doi.org/10.2118/SPE-172912-MS.

Droste, H.H.J., 1997. Stratigraphy of Lower Paleozoic Haima Supergroup of Oman: GeoArabia2: 419-492.

Gomaa, Ahmed M., Al Salmi, Hussein, and Casero, Alberto. "Developments In Efforts to Replace Borate Crosslinked with High Viscosity Friction Reducers. Part I: Laboratory Tests." Paper presented at the SPE Internation Hydraulic Fracturing Technical Conference, Muscat, Sultanate of Oman, September 2025. SPE-226629.

Pollastro, Richard M. Ghaba Salt Basin Province and Fahud Salt Basin Province, Oman: Geological Overview and Total Petroleum Systems. US Department of the Interior, US Geological Survey, 1999.

Rylance, M, Nicolaysen, A, Ishteiwy, O. 2011. Hydraulic Fracturing: Key to Effective Khazzan-Makarem Tight-Gas Appraisal. Paper presented at the SPE Middle East Unconventional Gas Conference and Exhibition, Muscat, Oman, 31 January–2 February. SPE-142783-MS. https://doi.org/10.2118/142783-MS.

Vitthal, S., Van Zanten, R., Malbrough, J., Tallin, A., Reynolds, M. A., Dawson, J., and K. Hoeman. "Development and Evaluation of an Alternative Fluid for Offshore Fracturing of Moderate to High Permeability Formations." Paper presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, September 2024. doi: https://doi.org/10.2118/220784-MS